

KUSH AGARWAL

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OBJECTIVE

Second-year Computer Science undergraduate specializing in AI/ML, with proven open-source contributions to Kornia and Gammapy and hands-on experience building deployed AI agents and full-stack ML applications. Seeking a Software Engineering / AI-ML internship to apply strong Python, deep learning, and system design skills in a fast-paced, product-driven environment.

EDUCATION

B.Tech in Computer Science & Engineering (Specialization: AI & ML) 2025 – 2029 GITAM University, Visakhapatnam, India

OPEN SOURCE CONTRIBUTIONS

- **Gammapy — Official Contributor and Co-Author of v2.1 Release** | [View PR](#) | [Citation](#)
Optimized core N-dimensional mapping algorithms using dynamic tensor slicing with NumPy, achieving a ~500x performance speedup for irregular geometry cropping.
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TECHNICAL PROJECTS

- **Word Embedding Model (PyTorch)** | [GitHub Repo](#)
Built a word embedding model from scratch in PyTorch (no pretrained APIs) and trained it on the WikiText-103 dataset to deepen understanding of embedding architectures.
 - **Agentic Chat Support System** | [Live on Hugging Face](#)
Built a LangChain-powered agentic support system with real tool-calling, multi-turn memory, and dynamic intent routing;
deployed live on Hugging Face Spaces.
 - **GitHub AI Agent** | [GitHub Repo](#)
Engineered an AI agent with LangGraph and MCP that autonomously traverses GitHub repositories, plans coding tasks, invokes GitHub tools to gather context, and verifies its own output before returning a final answer.
 - **Chalkie** | [Live](#) | [Github Repo](#)
Chalkie is an RAG based AI model which can explain complex workflows and concepts via interactive visualizations, it is a sync of three states together that are whiteboard visuals , voice over and a laser pointer syncing both together.
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TECHNICAL SKILLS

- **Languages:** Python, C, C++, TypeScript and SQL
- **Libraries/Frameworks:** PyTorch, NumPy, Pandas, Flask, spaCy (NLP), LangChain and LangGraph
- **Tools & Workflows:** Git, GitHub, CI/CD, Hugging Face Spaces, Kaggle
- **System design**